

H2

ANDERSON JUNIOR COLLEGE
HIGHER 2

CANDIDATE
NAME

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PDG
INDEX NUMBER

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BIOLOGY 9744/02

Paper 2 Structured Questions

12 September 2017
Tuesday

2 hours

Additional Materials: Answer Paper

READ THESE INSTRUCTIONS FIRST

Write your name and PD group on all the work you hand in.
Write in dark blue or black pen.
You may use a soft pencil for any diagrams, graph or rough working.
Do not use paper clips, highlighters, glue or correction fluid.

Answer **all** questions.

All working for numerical answers must be shown.
At the end of the examination, fasten all your work securely together.
The number of marks is given in brackets [] at the end of each question or part question.

Calculators may be used

For Examiner's Use	
1	
2	
3	
4	
5	
6	
7	
8	
Total	100

This document consists of **22** printed pages.

Answer **all** the questions.

- 1** Lactose intolerance in humans is the inability to hydrolyse lactose due to the lack of the enzyme lactase in the alimentary canal. As a result, bacteria in the large intestines feed on the lactose and produces fatty acids and methane which lead to diarrhoea and flatulence.

Bacteria have been used to produce lactase (Fig. 1.1) on an industrial scale as a dietary supplement for people who are lactose intolerant. Human lactase consists of a single 160 kDa polypeptide chain that localizes to the brush border membrane of intestinal epithelial cells.

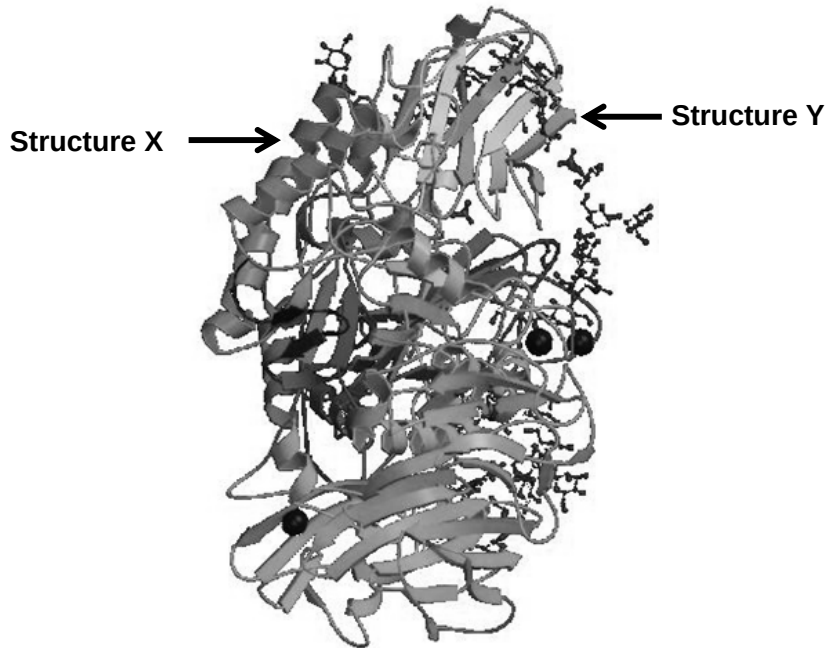


Fig. 1.1

- (a) With reference to the **Fig. 1.1**,
 (i) state the levels of organization seen in the structure of lactase.

[1]

- (ii) Describe structures **X** and **Y**.

[2]

(b) Fig. 1.2 shows the hydrolysis of lactose.

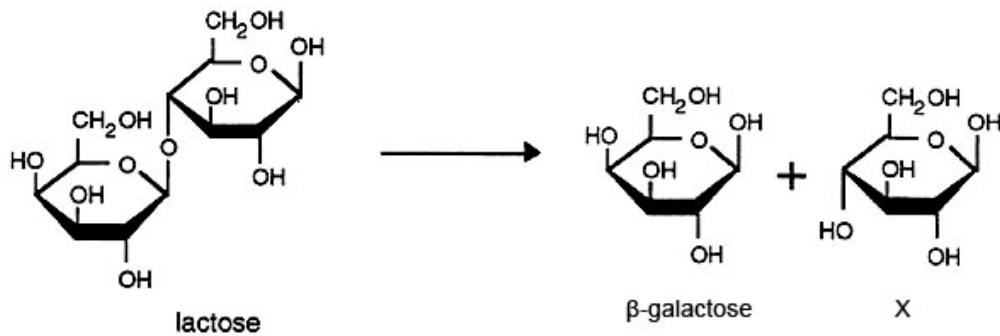


Fig. 1.2

Describe the hydrolysis of lactose, naming the bond that is broken and product X.

[2]

(c) Lactase and lactose are protein and carbohydrates respectively.
Explain why there are fewer types of carbohydrate polymers compared to protein polymers.

[3]

Cell organelles can be separated by centrifuging a cell extract in a sucrose density gradient. The organelles settle at the level in the sucrose solution which has the same density as their own.

The cells used to synthesize lactase were lysed and the cell extract centrifuged in a sucrose density gradient. Three distinct fractions of nuclei, mitochondria and ribosomes (in no particular order) were obtained. The three fractions **A**, **B** and **C** are shown in the **Fig. 1.3**.

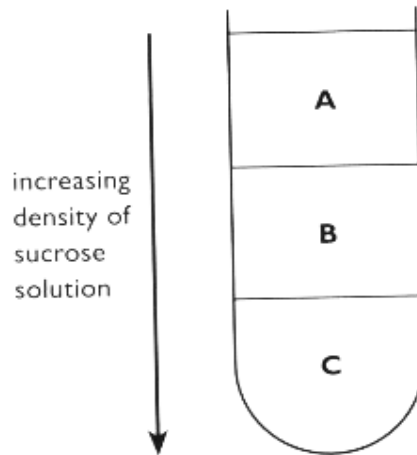


Fig. 1.3.

- (d)** Identify the organelle in each fraction and describe its role in the synthesis of lactase.

[4]

[Total: 12 marks]

- 2 Recently, scientists discovered the presence of a population of bone-marrow derived stem cells that have the ability to form heart muscles cells when transferred to the heart. The stem cells were removed from the bone marrow and cultured so that they divided by mitosis. It was proposed that these stem cells resembled embryonic stem cells.

(a) (i) Describe **two** similarities between these bone-marrow derived stem cells and embryonic stem cells.

[2]

(ii) Describe how the rate of mitosis is controlled.

[4]

(iii) State an advantage of using bone marrow derived stem cells rather than heart stem cells for the treatment of heart diseases.

[1]

- (c) Troponin is a protein that is integral to muscle contraction in heart muscles. **Fig. 2.1** shows part of its DNA sequence. The entire sequence is 63 base pairs.

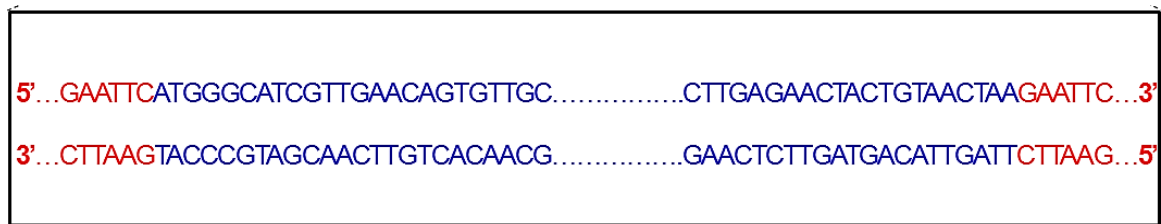


Fig. 2.1

PCR can be used to confirm presence of troponin DNA sequence. The following pair of primers are used.

Primer	Primer sequence
1	5' AATTCATGGGCATCG 3'
2	5' GAATTCTTAGTTACA 3'

- (i) In the boxed area in Fig. 2.1, circle and label the DNA sequences where Primers 1 and 2 will anneal. [1]

- (ii) Explain how results of gel electrophoresis of the PCR products are able to show that troponin DNA has been successfully amplified.

..... [1]

- (iii) Besides the use of PCR, nucleic acid hybridisation can also be used to determine presence of troponin DNA.

Outline how nucleic acid hybridisation can be used to identify troponin DNA.

.....

.....

.....

.....

.....

..... [4]

[Total: 13 marks]

- 3 The pie chart in Fig. 3.1 shows the relative length of time of each of the stages that occur during a particular eukaryotic cell cycle. This complete cycle takes 15 hours.

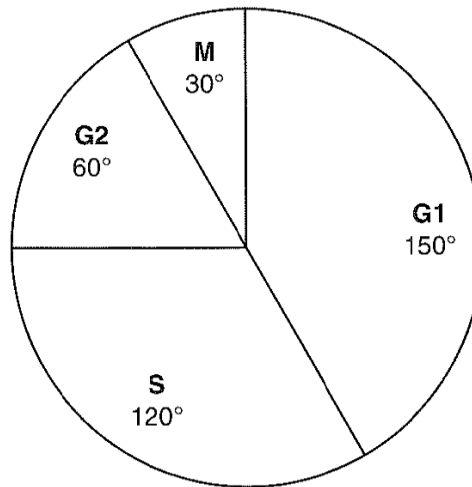


Fig. 3.1

- (a) Using the information provided above, calculate how long interphase lasts.

You will lose marks if you do not show your working or do not use appropriate units.

[2]

- (b) During the cell cycle there are a number of checkpoints.

State one function of these checkpoints and explain what might occur as a result of dysregulation of these checkpoints.

[3]

The diagram illustrates the progression of colorectal cancer through a series of genetic alterations and corresponding stages of tumor development. The progression is shown as a sequence of six stages, each represented by a cross-section of the intestinal wall. The stages are: Normal epithelium, Early adenoma, Intermediate adenoma, Late adenoma, Carcinoma, and Metastasis. The genetic alterations are listed above the stages, with arrows indicating the progression from one stage to the next. The alterations are: Wnt pathway activation (affecting APC), KRAS activation, TGF-beta response inactivation (affecting Smad2/4), Loss of p53 function (affecting p53), and Other genetic alterations. The progression is shown as a sequence of six stages, each represented by a cross-section of the intestinal wall. The stages are: Normal epithelium, Early adenoma, Intermediate adenoma, Late adenoma, Carcinoma, and Metastasis. The genetic alterations are listed above the stages, with arrows indicating the progression from one stage to the next. The alterations are: Wnt pathway activation (affecting APC), KRAS activation, TGF-beta response inactivation (affecting Smad2/4), Loss of p53 function (affecting p53), and Other genetic alterations.

Affected event	Wnt pathway activation	EFGR signalling activation	TGF-beta response inactivation	Loss of p53 function	Other genetic alterations
Gene(s) involved	APC	KRAS	Smad2/4	p53	
Normal epithelium	Early adenoma	Intermediate adenoma	Late adenoma	Carcinoma	Metastasis

(i) Using the Fig. 3.2, explain why the development of cancer is a multi-step process.

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- (ii) The majority of all colorectal cancers occur sporadically without any known cause, but certain groups of people have a predisposition to the development of cancer of the large intestine. These people may carry specific genetic mutations or have relatives with the condition.

Approximately 15% of all colorectal cancer cases are familial, with the most common inherited conditions being familial adenomatous polyposis (FAP). Patients with FAP have a lifetime risk of the development of colon cancer that approaches 100%. Patients with FAP have a germline inactivation of one APC allele. Adenoma formation is faster, ~~but~~ progression from adenoma to carcinoma has the same rate as sporadic colorectal cancer as shown in Fig. 3.3.

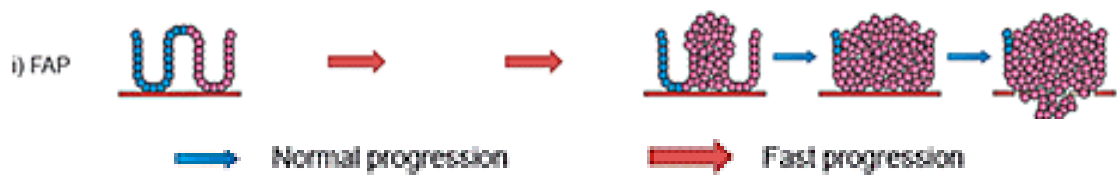


Fig. 3.3

Using the information above and your own understanding of the development of cancer, suggest why patients with FAP form adenoma faster.

[2]

[Total: 12 marks]

- 4 Genome editing is the process in which a DNA target sequence is replaced by a desired sequence. **Fig. 4.1** shows how the process is being done by Cas9 enzyme which makes use of a guide RNA to achieve the editing effect. This can be done in embryos so that those children who are born from parents with the genetic disease alleles would not suffer from the genetic disease.

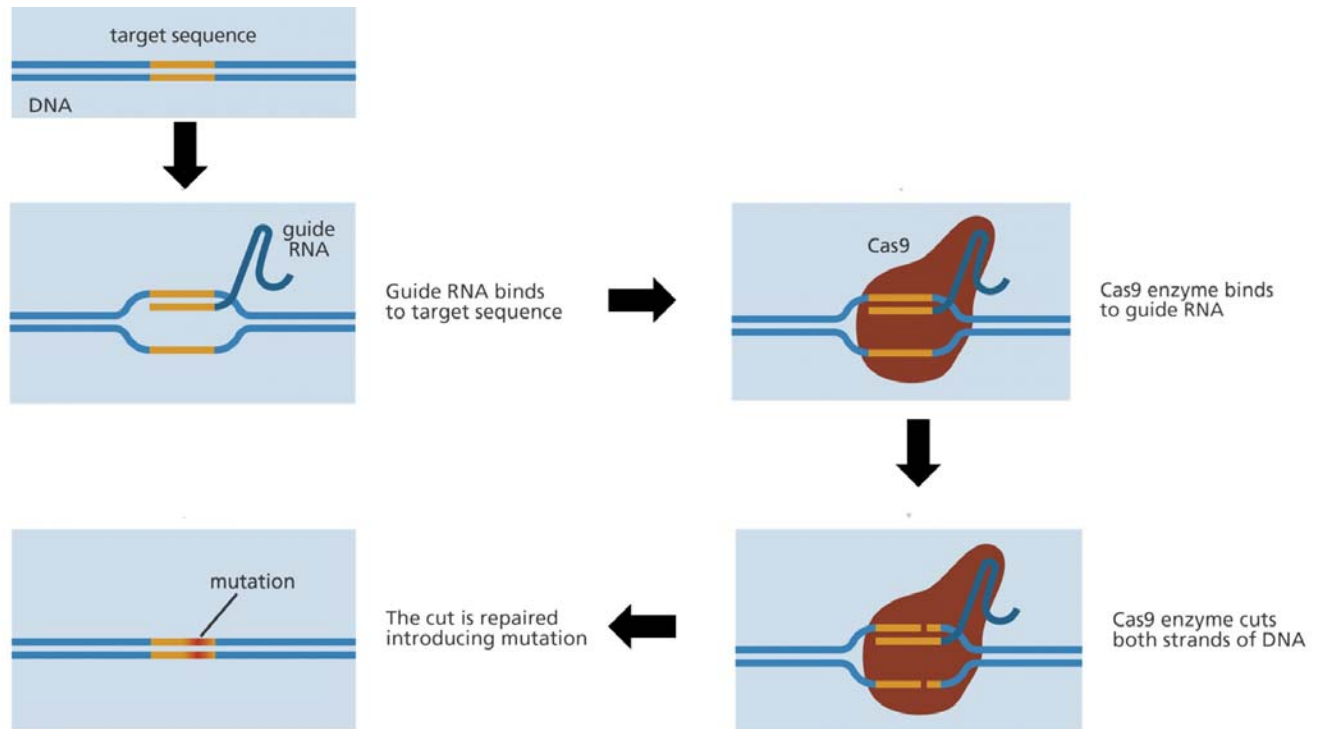


Fig. 4.1

- (a) With reference to Fig. 4.1, describe how the guide RNA and Cas9 enzyme are used to cut both strands of DNA.

[3]

- (b) (i) Explain why gene editing done on embryos help to prevent children born from suffering from the genetic disease.

[2]

- (ii) Suggest and explain if the mutation introduced by gene editing as shown in Fig. 4.1 should be dominant or recessive.

.....
..... [2]

- (c) Describe how mutations in DNA arise in nature.

.....
..... [2]

RNA plays a very important role in many biological processes. One of them is transfer RNA (tRNA) which has extensive intramolecular hydrogen bonds.

- (d) (i) State **two** importance of having such bonds.

.....
..... [2]

- (ii) Relate the structure of tRNA to its functions.

.....
.....
..... [2]

[Total: 13 marks]

- 5 Fig. 5.1 represents a bacteria DNA and a eukaryotic chromosome in metaphase of mitosis, not drawn to scale.



Fig. 5.1

- (a) State **two** ways in which the organization of genes found in these two structures differ and suggest **one** advantage of this to the bacterium.

[3]

- (b) In 1946, Joshua Lederberg and Edward Tatum proposed that bacterial cells undergo genetic recombination. To test their hypothesis, they conducted experiments using two bacteria strains of *Escherichia coli* (*E.coli*), A and B, with different nutritional requirements.

Strain A, B and a mixture of both strains were grown on culture plates containing minimal medium that does not contain essential amino acids. The results are shown in Fig. 5.2.

Mutant genes (–) do not code for enzymes that synthesize amino acids. Note that all five amino acids are required for bacterial growth.

Bacterial strains	Genes for biosynthesis of amino acids	Mutant genes for biosynthesis of amino acids
A	thr ⁺ leu ⁺ thi ⁺	met [–] bio [–]
B	met ⁺ bio ⁺	thr [–] leu [–] thi [–]

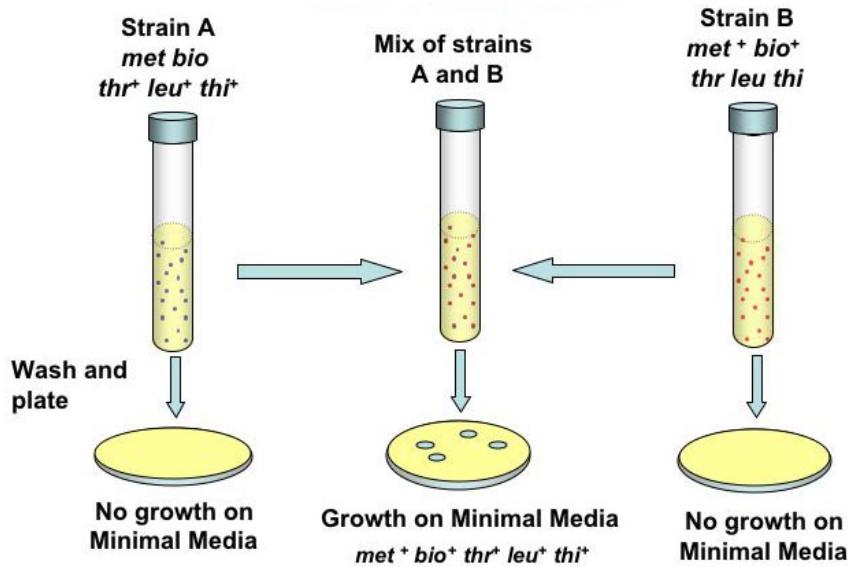


Fig. 5.2

Another researcher, Bernard Davis also worked with the same hypothesis. In his experiment he constructed a U-tube in which the two arms were separated by a fine filter. The pores of the filter were too small to allow bacteria to pass through but large enough to allow easy passage of the fluid medium, any dissolved substances and free DNA. The results are shown in Fig. 5.3.

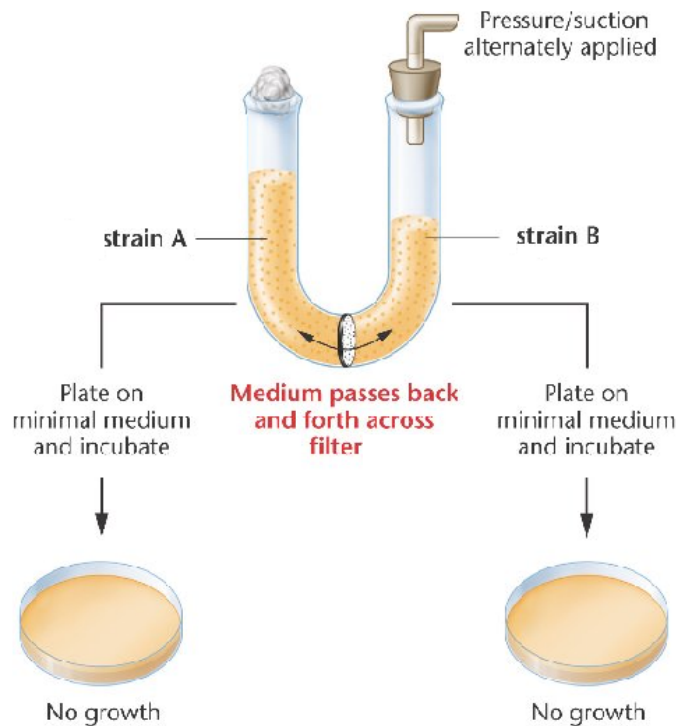


Fig. 5.3

- (i) Using the results of the two experiments shown in Fig. 5.2 and Fig. 5.3 and your understanding of genetic recombination in bacteria, state the genetic recombination that has taken place between Strain A and B. Explain your answer.

[6]

- (c) In 2016, a pathogenic strain of *E.coli* found on unwashed salad caused food poisoning in 151 people in Britain, leaving two of them dead.

Describe how such pathogens are usually treated using a named example.

[3]

[Total: 12 m]

- 6 In anaerobic respiration in yeast, the pyruvate molecules are broken down to produce ethanol and carbon dioxide. The release of carbon dioxide can be used to investigate the rate of anaerobic respiration.

Fig. 6.1 shows an experiment which was set up to find the rate of anaerobic respiration.

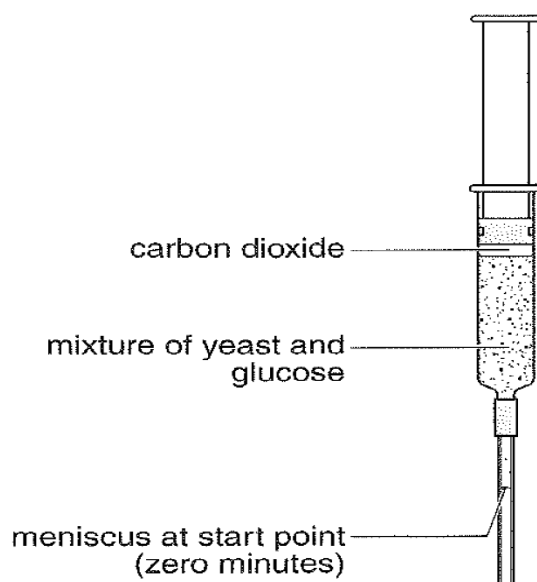


Fig. 6.1

The meniscus moves down the tube as carbon dioxide is released.

Table 6.1 shows the distance moved by the meniscus from the start point. This was recorded every 10 minutes.

Table 6.1

Time/ min	0	10	20	30	40	50	60	70	80	90
Distance travelled by meniscus from start point/ mm	0	1	2	5	9	14	21	45	73	98

- (a)** The rate of anaerobic respiration can be calculated by using the rate of movement of the meniscus.

Calculate the rate of anaerobic respiration between 70 and 80 minutes.

You will lose marks if you do not show your working.

(b) This experiment was repeated three more times. Each time, the glucose (a monosaccharide) was replaced with a different disaccharide sugar:

- Maltose – a disaccharide of glucose and glucose
- Sucrose – a disaccharide of glucose and fructose
- Lactose – a disaccharide of glucose and galactose.

Tables 6.2 (a), (b) and (c) show the results of these experiments.

Table 6.2 (a): Using maltose

Time/ min	0	10	20	30	40	50	60	70	80	90
Distance travelled by meniscus from start point/ mm	0	0	0	0	0	2	3	6	9	12

Table 6.2 (b): Using sucrose

Time/ min	0	10	20	30	40	50	60	70	80	90
Distance travelled by meniscus from start point/ mm	0	0	0	1	3	11	22	37	48	61

Table 6.2 (c): Using lactose

Time/ min	0	10	20	30	40	50	60	70	80	90
Distance travelled by meniscus from start point/ mm	0	0	0	0	0	0	0	0	0	0

With reference to the information provided in Tables 6.2 (a), (b) and (c) and your biological knowledge:

(i) Describe the difference in the results for maltose and sucrose, and suggest one explanation for this difference,

- (ii) Suggest two explanations for the results for lactose.

[2]

- (c) An electron micrograph of yeast, *Candida albicans*, is shown in Fig. 6.2.

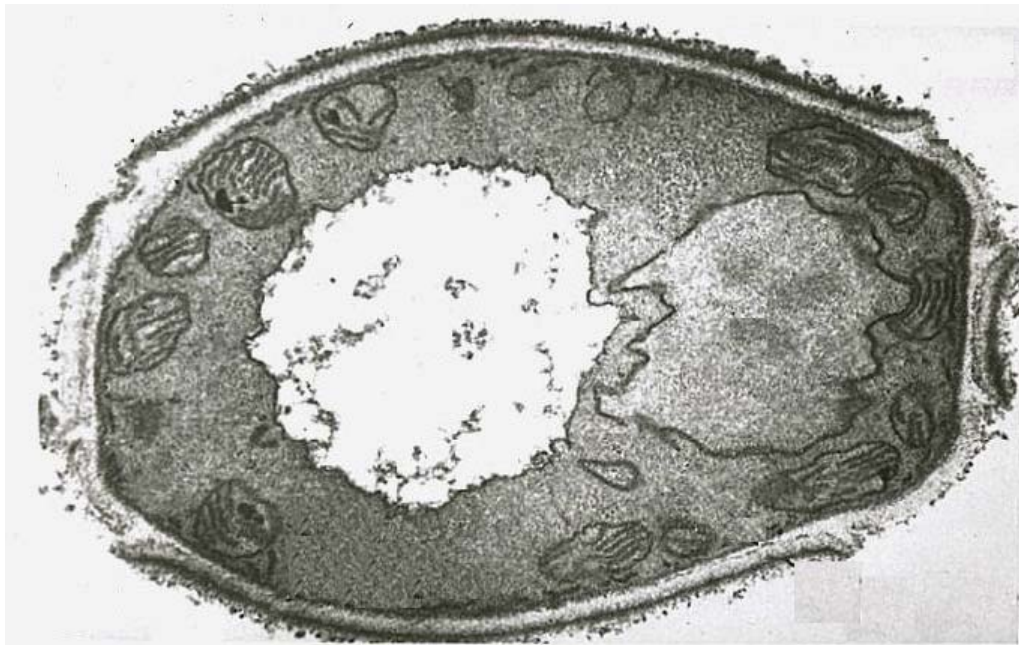


Fig. 6.2

- (i) On Fig. 6.2, label site of
- Glycolysis
 - Oxidative phosphorylation
- (ii) State one visible structure of mitochondria from Fig. 6.2 and describe how it supports mitochondria's function.

[2]

[1]

(ii) Besides location, compare between oxidative phosphorylation and photophosphorylation.

[illegible]

[4]

[Total: 13 marks]

- 7 Pigment production in onions is controlled by two enzymes resulting in three different coloured bulbs - red, yellow and white.

A pure-white strain crossed with a pure-red strain produces an all-white offspring (F_1). Two F_1 onions with white bulbs were crossed. The F_2 generation was found to consist of 2170 white, 530 red and 180 yellow-bulbs.

The alleles are represented by the following symbols:

I: no production of pigment	i: production of pigment
R: red pigment	r: yellow pigment

- (a) State the mode of inheritance in the onion bulb colour.

.....[1]

- (b) Explain the results of the cross by drawing a genetic diagram in the space below.

[5]

- (c) Explain how different genotypes give rise to different phenotypes.

[4]

- (e) Suggest how a farmer may determine if a red onion is homozygous in both loci.

[2]

[Total: 12 marks]

- 8 Antibodies against tuberculosis are produced by plasma cells during an immune response.

Fig. 8.1 shows a diagram of an antibody molecule.

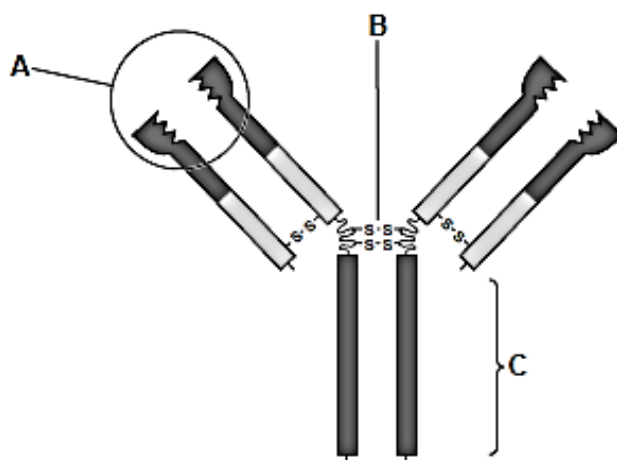


Fig. 8.1

- (a) Explain the functions of the parts labelled A, B and C.

(i) A

[2]

(ii) B

[1]

(iii) C

[1]

(b) Explain why tuberculosis (TB) is known as an infectious disease.

[3]

(c) Outline the roles of antibiotics in the treatment of infectious diseases, such as TB.

[3]

(d) While TB is a bacterial infectious disease, HIV is a viral infectious disease.

Explain how HIV cause diseases in humans through the disruption of host tissue and functions.

[3]

[Total: 13 marks]